

- 2 Stem cells are found throughout the human body. Lgr5⁺ stem cells are found in the lining of the small intestine.

Fig. 2.1 is a flow chart showing stages in the development of one of the daughter cells produced by the mitotic division of an Lgr5⁺ stem cell.

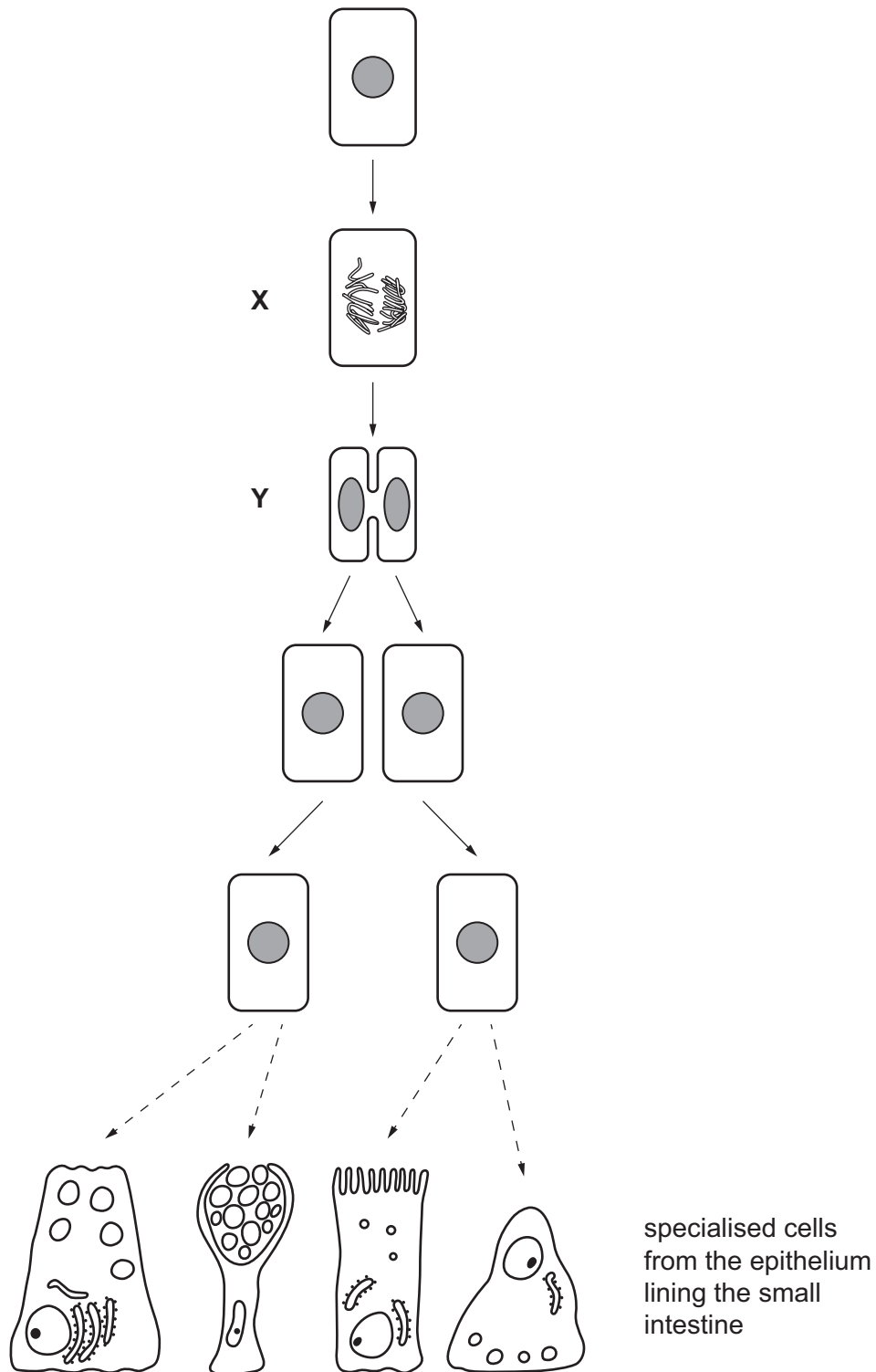


Fig. 2.1



- (a) (i) Explain why stem cells are required in places such as the lining of the small intestine.

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.....

..... [2]

- (ii) Name the stage of mitosis shown in cell **X** in Fig. 2.1.

..... [1]

- (iii) State the part of the cell cycle shown at **Y** in Fig. 2.1.

..... [1]

- (b) Explain the role of centromeres in the cell cycle.

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..... [2]

(c) Fig. 2.2 shows three types of specialised cell that develop from Lgr5⁺ stem cells in the small intestine.

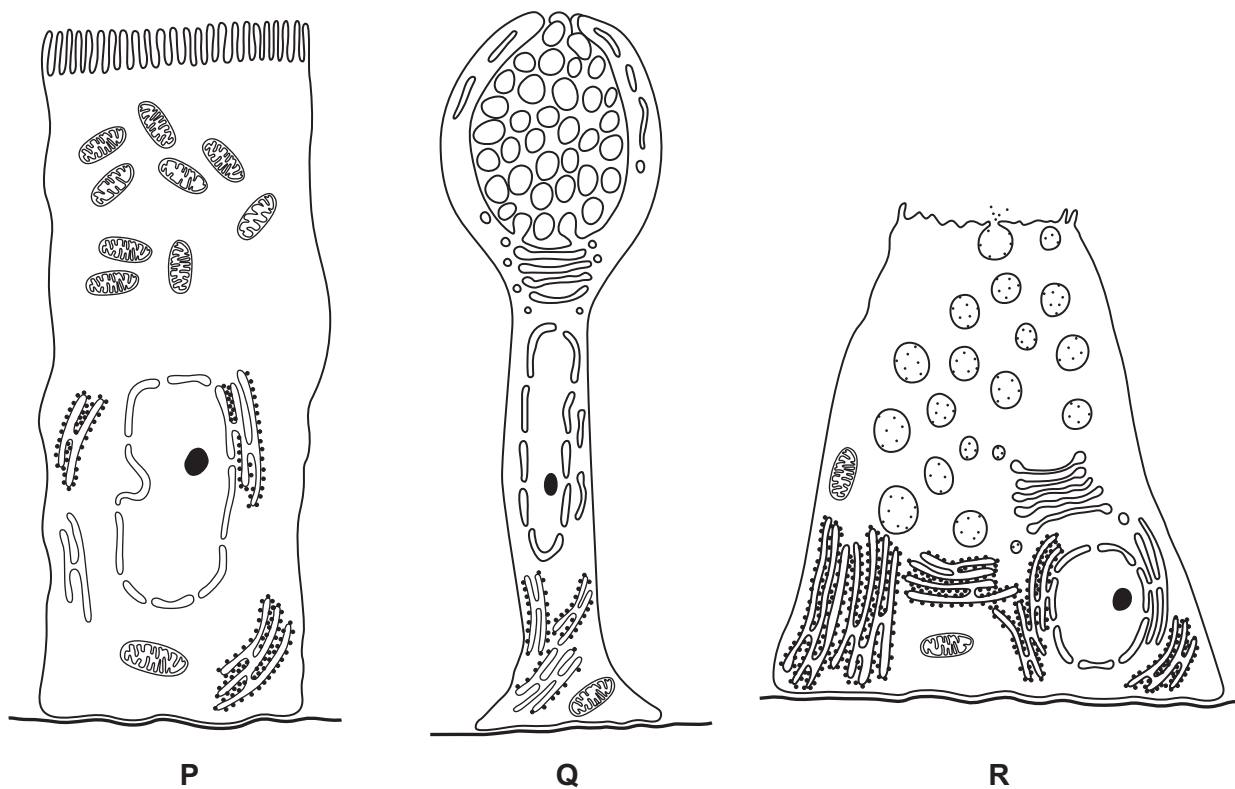


Fig. 2.2

The structural features of a cell indicate its likely function.

Suggest a function of each of the cells shown in Fig. 2.2, **and** explain how the structure of each cell supports your suggestion.

cell **P** function

explanation

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cell **Q** function

explanation

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cell **R** function

explanation

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[6]

[Total: 12]